

UNIT-1

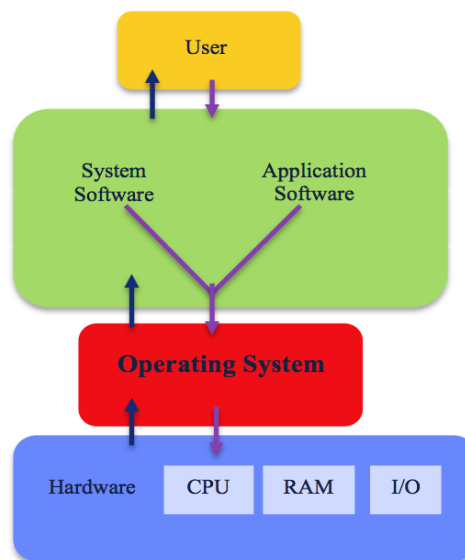
INTRODUCTION AND SYSTEM STRUCTURE:

Introduction: Operating system, functions of operating system, types of operating system, computer system organization, computer system architecture, operating system structure, operating system operations, computing environments, open-source operating systems.

Operating System Structures: operating system services, system calls, types of system calls, system programs, operating system structure, operating system debugging and system boot

What is operating system:

- An Operating System (OS) is an interface between a computer user and computer hardware.
- An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers.
- Some popular Operating Systems include Linux Operating System, Windows Operating System, VMS, OS/400, AIX, z/OS, etc.



Functions of operating system:

1. Memory management
2. Process management
3. Device management
4. File management
5. Security
6. Control over the performance
7. Job counting

Memory Management:

Memory management refers to management of Primary Memory or Main Memory. Main memory is a large array of words or bytes where each word or byte has its own address. Main memory provides

B. Keerthana, Ass.Prof, CSE department

a fast storage that can be accessed directly by the CPU. For a program to be executed, it must in the main memory.

An Operating System does the following activities for memory management –

- Keeps tracks of primary memory, i.e., what part of it are in use by whom, what part are not in use.
- In multiprogramming, the OS decides which process will get memory when and how much.
- Allocates the memory when a process requests it to do so.
- De-allocates the memory when a process no longer needs it or has been terminated.

Processor Management

In multiprogramming environment, the OS decides which process gets the processor when and for how much time. This function is called **process scheduling**. An Operating System does the following activities for processor management –

- Keeps tracks of processor and status of process. The program responsible for this task is known as **traffic controller**.
- Allocates the processor (CPU) to a process.
- De-allocates processor when a process is no longer required.

Device Management

An Operating System manages device communication via their respective drivers. It does the following activities for device management –

- Keeps tracks of all devices. Program responsible for this task is known as the **I/O controller**.
- Decides which process gets the device when and for how much time.
- Allocates the device in the efficient way.
- De-allocates devices.

File Management

A file system is normally organized into directories for easy navigation and usage. These directories may contain files and other directions.

An Operating System does the following activities for file management –

- Keeps track of information, location, uses, status etc. The collective facilities are often known as **file system**.
- Decides who gets the resources.
- Allocates the resources.
- De-allocates the resources.

Security – By means of password and similar other techniques, it prevents unauthorized access to programs and data.

Control over system performance – Recording delays between request for a service and response from the system.

Job accounting – Keeping track of time and resources used by various jobs and users.

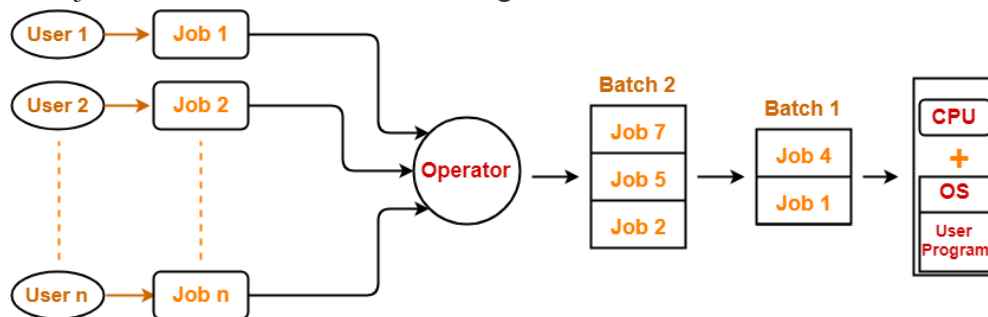
Error detecting aids – Production of dumps, traces, error messages, and other debugging and error detecting aids.

Coordination between other softwares and users – Coordination and assignment of compilers, interpreters, assemblers and other software to the various users of the computer systems.

Type of operating system:

1. Batch operating system: In batch operating system

- Firstly, user prepares his job using punch cards.
- Then, he submits the job to the computer operator.
- Operator collects the jobs from different users and sort the jobs into batches with similar needs.
- Then, operator submits the batches to the processor one by one.
- All the jobs of one batch are executed together.



Batch Operating System

Advantages-

- It saves the time that was being wasted earlier for each individual process in context switching from one environment to another environment.
- No manual intervention is needed.

disadvantages:

→ Priority can not be set for the jobs:

In batch operating system,

- All the jobs of a batch are executed sequentially one after the other.
- The output is obtained only after all the jobs are executed.
- Thus, priority can not be implemented if a certain job has to be executed on an urgent basis.

→ Batch operating system may lead to starvation.

In batch operating system,

- The jobs of a particular batch might take long time for their execution.
- This might lead to starvation to other jobs in other batches.

→ CPU may remain idle for a long time.

- If the jobs of a batch require some I/O operation, then CPU must wait till the I/O operation gets completed.
- Since I/O devices are very slow, CPU remains idle for a long time.
- CPU can not take any other job and execute it.

→ There is a lack of interaction between a user and his job.

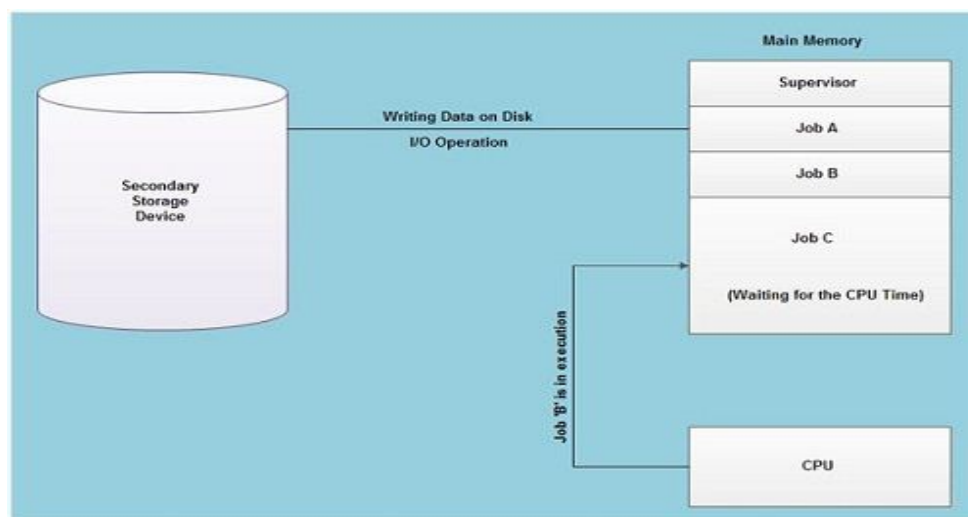
- Once a batch is submitted for execution, the user is not able to interact with any of his jobs.

- If a job requires the user to input data during run time, then user must wait till the other jobs of the batch get executed.
- This also increases the overall execution time.

2. Multiprogramming operating system:

To overcome the problem of underutilization of CPU and main memory, the multi-programming was introduced. The multi-programming is interleaved execution of multiple jobs by the same computer.

In multi-programming system, when one program is waiting for I/O transfer; there is another program ready to utilize the CPU. So it is possible for several jobs to share the time of the CPU. But it is important to note that multi-programming is not defined to be the execution of jobs at the same instance of time. Rather it does mean that there are a number of jobs available to the CPU (placed in main memory) and a portion of one is executed then a segment of another and so on. A simple process of multi-programming is shown in figure



As shown in fig, at the particular situation, job 'A' is not utilizing the CPU time because it is busy in I/O operations. Hence the CPU becomes busy to execute the job 'B'. Another job C is waiting for the CPU for getting its execution time. So in this state the CPU will never be idle and utilizes maximum of its time.

A program in execution is called a "Process", "Job" or a "Task". The concurrent execution of programs improves the utilization of system resources and enhances the system throughput as compared to batch and serial processing. In this system, when a process requests some I/O to allocate; meanwhile the CPU time is assigned to another ready process. So, here when a process is switched to an I/O operation, the CPU is not set idle.

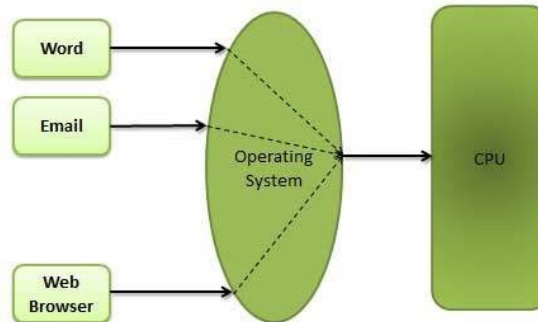
3. Multi tasking / time sharing operating system:

Multitasking is when multiple jobs are executed by the CPU simultaneously by switching between them. Switches occur so frequently that the users may interact with each program while it is running.

An OS does the following activities related to multitasking –

- The user gives instructions to the operating system or to a program directly, and receives an immediate response.
- The OS handles multitasking in the way that it can handle multiple operations/executes multiple programs at a time.

- Multitasking Operating Systems are also known as Time-sharing systems.
- These Operating Systems were developed to provide interactive use of a computer system at a reasonable cost.
- A time-shared operating system uses the concept of CPU scheduling and multiprogramming to provide each user with a small portion of a time-shared CPU.
- Each user has at least one separate program in memory.



- A program that is loaded into memory and is executing is commonly referred to as a **process**.
- When a process executes, it typically executes for only a very short time before it either finishes or needs to perform I/O.
- Since interactive I/O typically runs at slower speeds, it may take a long time to complete. During this time, a CPU can be utilized by another process.
- The operating system allows the users to share the computer simultaneously. Since each action or command in a time-shared system tends to be short, only a little CPU time is needed for each user.
- As the system switches CPU rapidly from one user/program to the next, each user is given the impression that he/she has his/her own CPU, whereas actually one CPU is being shared among many users.

4. Distributed operating system:

A distributed operating system (**DOS**) is an essential type of operating system. Distributed systems use many central processors to serve multiple real-time applications and users. As a result, data processing jobs are distributed between the processors.

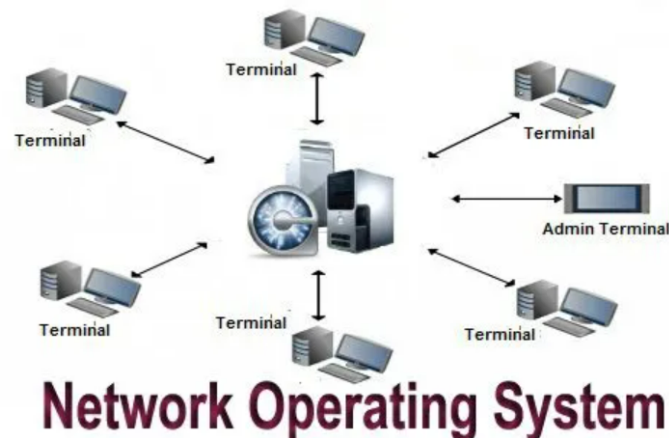
It connects multiple computers via a single communication channel. Furthermore, each of these systems has its own processor and memory. Additionally, these **CPUs** communicate via high-speed buses or telephone lines. Individual systems that communicate via a single channel are regarded as a single entity. They're also known as **loosely coupled systems**. This operating system consists of numerous computers, nodes, and sites joined together via **LAN/WAN**.



5. Network operating system:

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Network operating system is like as software that is installed on the server side on the network infrastructure. Network operating system provides the several functionalities for managing the data, applications, security, and other functions. In this system, all computers and other terminal are connected each other through LAN (Local Area Network) line or Inter-network system. Main objective of NOS (Network operating system) is to share data, printer access, and other devices. Novell NetWare was introduced as a first network operating system, in 1983.



6. Real time operating system:

Real-time operating system (RTOS) is an operating system intended to serve real time application that process data as it comes in, mostly without buffer delay. The full form of RTOS is Real time operating system.

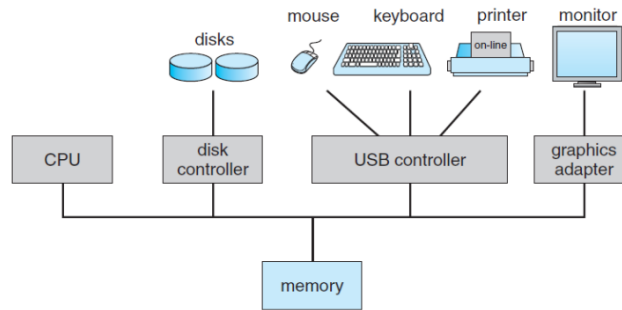
In a RTOS, Processing time requirement are calculated in tenths of seconds increments of time. It is time-bound system that can be defined as fixed time constraints. In this type of system, processing must be done inside the specified constraints. Otherwise, the system will fail.

Here are important features of RTOS:

- Occupy very less memory
- Consume fewer resources
- Response times are highly predictable
- Unpredictable environment
- The Kernel saves the state of the interrupted task and then determines which task it should run next.
- The Kernel restores the state of the task and passes control of the CPU for that task.

Computer System Organization:-

Computer System Organization provides general knowledge of the structure of a computer system. Computer System Operation: A modern general-purpose computer system consists of one or more CPU's and several device controllers connected through a common bus that provides access to shared memory as shown below.



The CPU and the device controllers can execute in parallel, competing for memory cycles. To ensure orderly access to the shared memory, a memory controller synchronizes access to the memory.

For a computer to start running—for instance, when it is powered up or rebooted—it needs to have an initial program to run. This initial program, or bootstrap program, tends to be simple. Typically, it is stored within the computer hardware in read-only memory (ROM) or electrically erasable programmable read-only memory (EEPROM), known by the general term firmware. It initializes all aspects of the system, from CPU registers to device controllers to memory contents. The bootstrap program must know how to load the operating system and how to start executing that system. To accomplish this goal, the bootstrap program must locate the operating-system kernel and load it into memory.

Once the kernel is loaded and executing, it can start providing services to the system and its users. Some services are provided outside of the kernel, by system programs that are loaded into memory at boot time to become system processes. On UNIX, the first system process is “init,” and it starts many other daemons. Once this phase is complete, the system is fully booted, and the system waits for some event to occur.

The occurrence of an event is usually signalled by an interrupt from either the hardware or the software. Hardware may trigger an interrupt at any time by sending a signal to the CPU, usually by way of the system bus. Software may trigger an interrupt by executing a special operation called a system call (also called a monitor call).

When the CPU is interrupted, it stops what it is doing and immediately transfers execution to a fixed location. The fixed location usually contains the starting address where the service routine for the interrupt is located. The interrupt service routine executes; on completion, the CPU resumes the interrupted computation.

Computer system architecture:

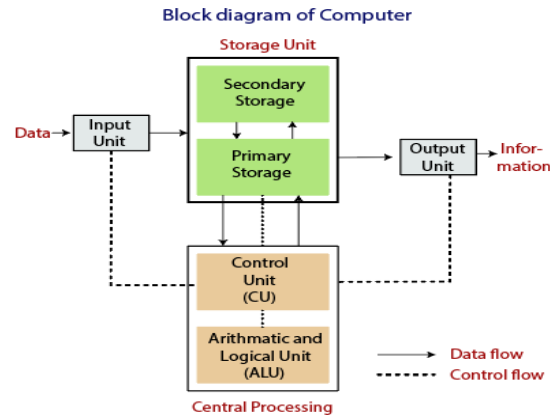
What is a Computer? : The computer is an electronic device which operates under the control of instructions stored in its memory. A computer can take data from the user through input devices (**Input**), process the user given data (**Processing**), produces the result to the user through output devices (**Output**) and stores data (**Information**) for future use.

A Computer can be defined as follows...

“The Computer is an electronic device which operates under the control of instructions stored in its memory and it takes the data from the user, a process that data gives the result and stores the result for future use.”

Block Diagram of a Computer:

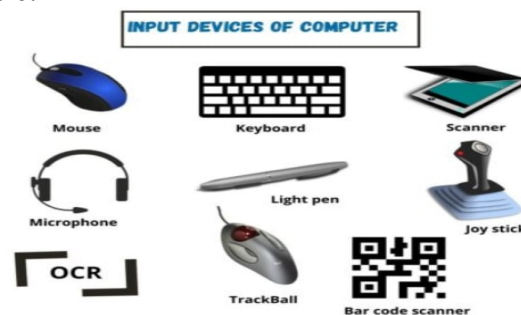
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Mainly the computer system consists of the following parts:

- Input devices
- Central Processing Unit (CPU)
- Storage unit/Memory unit
- Output device

1. **Input devices:** The input devices are used to give the data to the computer by user. Some of the input devices are:



2. **Output devices:** The output devices are used to display the result on computer to the user.



3. **Central Processing Unit (CPU):** The CPU is like the heart/brain of the computer. CPU is responsible for processing all the instructions which are given by the user to the computer system. The CPU is divided into two parts:

- **Control Unit (CU):** The control unit (CU) controls all the activities or operations which are performed inside the computer system. It receives instructions or information directly from the main memory of the computer.
- **Arithmetic Logic Unit (ALU):** This unit performs Arithmetic and Logical operations.

4. **Memory /Storage Unit:**

- Memory unit is used to store the large amount data with the help of “Primary Storage

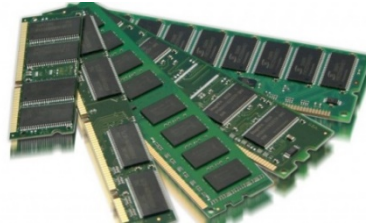
and Secondary Storage”.

- **Primary Storage:** Primary memory is also known as “main memory” which is located in the mother board of system. This is further divided into two types:

1) RAM 2) ROM

RAM (Read Only Memory):

- RAM stands for “Random Access Memory” which stores information temporarily i.e., it stores data which is currently used by the CPU.
- It is “Volatile Memory” because RAM needs power to work, when the computer power is turned off then all the data in it will be erased automatically.
- It allows both “read – write” operations and the storage capacity of RAM is 64MB to 16GB.



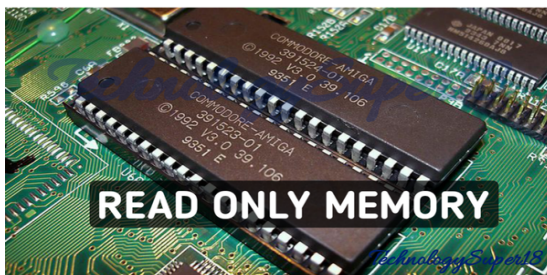
Types of RAM

- RAM can be classified into two types
 - **SRAM:** Static Random Access Memory
 - **DRAM:** Dynamic Random Access Memory.

Static RAM	Dynamic RAM
➤ SRAM uses transistor to store a single bit of data	➤ DRAM uses a separate capacitor to store each bit of data
➤ SRAM does not need periodic refreshment to maintain data	➤ DRAM needs periodic refreshment to maintain the charge in the capacitors for data
➤ SRAM's structure is complex than DRAM	➤ DRAM's structure is simplex than SRAM
➤ SRAM are expensive as compared to DRAM	➤ DRAM's are less expensive as compared to SRAM
➤ SRAM are faster than DRAM	➤ DRAM's are slower than SRAM
➤ SRAM are used in Cache memory	➤ DRAM are used in Main memory

ROM (Read Only Memory):

- ROM stands for “Read Only Memory” which only read the data and stores this data permanently.
- It is “Non Volatile” memory because the data cannot be erased even the computer power is turned off.
- ROM memory cannot be directly accessed by the CPU. The data is first transferred to RAM and then CPU is accessed from the RAM.



Types of ROM:

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- ROM can be classified into three types:
 - **PROM: “Programmable Read Only Memory”**. It is modified only once by a user.

Example: CD - R

- **EPROM: “Erasable and Programmable Read Only Memory”**. The content of this ROM can be erased by Ultra Violet rays and ROM can be reprogrammed.

Example: CD(R/W)

- **EEPROM: “Electrically Erasable and Programmable Read Only Memory”**. It can be erased electrically and reprogrammed about 10 thousand times.

Example: pen-drive, memory card.

Differences of ROM:

Types of ROM		
• PROM: – Programmable ROM – User can store programs only once. – User can make micro code program can be made that are needed mostly. – The process of making program in PROM is called ‘Burning’. – Example: CD-R	• EPROM: – Erasable PROM – Information can be removed by ultra violet rays. – Information can be re-write after removing previous information. – It is cheaper than PROM because it is re-useable. – Example: CD(RW)	• EEPROM: ◦ Electrically Erasable PROM ◦ Information can be removed by electric signals. ◦ It is the simplest way to store info in ROM. ◦ Now it is used to store BIOS in Memory. ◦ Example :Pen Drive

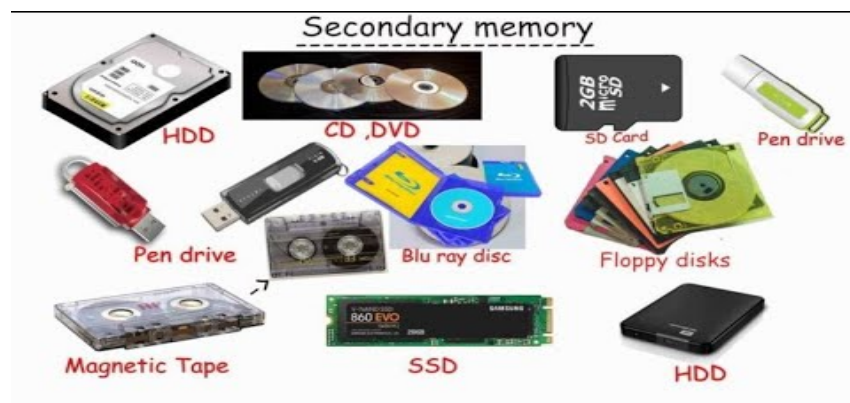
Cache Memory: is a semi-conductor (buffer)memory which lies between main memory and processor. It is mainly used for increasing the speed of RAM.

Difference b/w RAM & ROM:

BASIS FOR COMPARISON	RAM	ROM
Basic	It is a read-write memory.	It is read only memory.
Use	Used to store the data that has to be currently processed by CPU temporarily.	It stores the instructions required during bootstrap of the computer.
Volatility	It is a volatile memory.	It is a nonvolatile memory.
Stands for	Random Access Memory.	Read Only Memory.
Modification	Data in RAM can be modified.	Data in ROM can not be modified.
Capacity	RAM sizes from 64 MB to 4GB.	ROM is comparatively smaller than RAM.
Cost	RAM is a costlier memory.	ROM is comparatively cheaper than RAM.
Type	Types of RAM are static RAM and dynamic RAM.	Types of ROM are PROM, EPROM, EEPROM.

Secondary storage:

- **Secondary Storage:** The place where we store our personal data in computer system is known as “**Secondary Memory**”. We can easily retrieve when the data is needed. It is non volatile in nature so that we cannot lose the data when power supply is off.



There are two methods for accessing the data from it :-

- **Sequential** – This is the method in which we search the data sequentially or line by line until you find the desired data.
E.g.. Magnetic tape, etc.
- **Direct** – This is the method in which computer can go directly to the information that the user wants.
E.g. Magnetic disk: hard disk, CD/DVD etc.

Operating system structure:

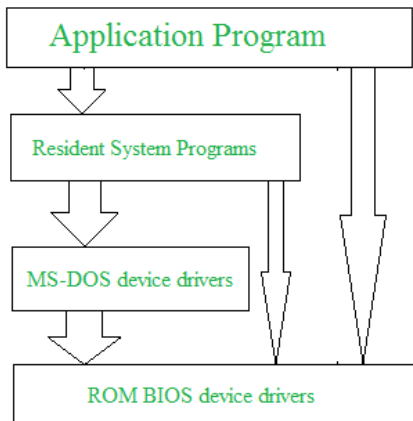
Operating system can be implemented with the help of various structures. The structure of the OS depends mainly on how the various common components of the operating system

are interconnected and melded into the kernel. Depending on this we have following structures of the operating system:

Simple structure:

Such operating systems do not have well defined structure and are small, simple and limited systems. The interfaces and levels of functionality are not well separated. MS-DOS is an example of such operating system. In MS-DOS application programs are able to access the basic I/O routines. These types of operating system cause the entire system to crash if one of the user programs fails.

Diagram of the structure of MS-DOS is shown below.



Advantages of Simple structure:

- It delivers better application performance because of the few interfaces between the application program and the hardware.
- Easy for kernel developers to develop such an operating system.

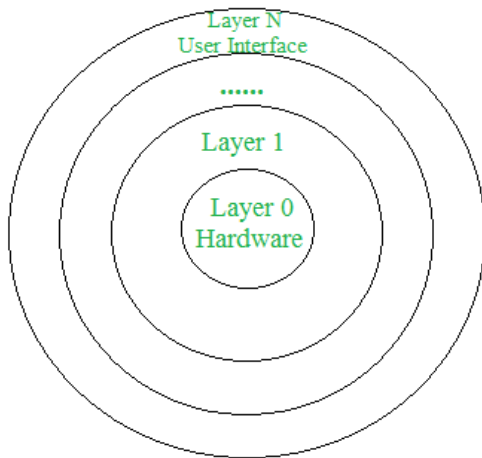
Disadvantages of Simple structure:

- The structure is very complicated as no clear boundaries exists between modules.
- It does not enforce data hiding in the operating system.

Layered structure:

An OS can be broken into pieces and retain much more control on system. In this structure the OS is broken into number of layers (levels). The bottom layer (layer 0) is the hardware and the topmost layer (layer N) is the user interface. These layers are so designed that each layer uses the functions of the lower level layers only. This simplifies the debugging process as if lower level layers are debugged and an error occurs during debugging then the error must be on that layer only as the lower level layers have already been debugged.

The main disadvantage of this structure is that at each layer, the data needs to be modified and passed on which adds overhead to the system. Moreover careful planning of the layers is necessary as a layer can use only lower level layers. UNIX is an example of this structure.



Advantages of Layered structure:

- Layering makes it easier to enhance the operating system as implementation of a layer can be changed easily without affecting the other layers.
- It is very easy to perform debugging and system verification.

Disadvantages of Layered structure:

- In this structure the application performance is degraded as compared to simple structure.
- It requires careful planning for designing the layers as higher layers use the functionalities of only the lower layers.

Micro-kernel:

This structure designs the operating system by removing all non-essential components from the kernel and implementing them as system and user programs. This results in a smaller kernel called the micro-kernel.

Advantages of this structure are that all new services need to be added to user space and does not require the kernel to be modified. Thus it is more secure and reliable as if a service fails then rest of the operating system remains untouched. Mac OS is an example of this type of OS.

Advantages of Micro-kernel structure:

- It makes the operating system portable to various platforms.
- As microkernels are small so these can be tested effectively.

Disadvantages of Micro-kernel structure:

- Increased level of inter module communication degrades system performance.

Monolithic Kernel:

A monolithic kernel is an operating system architecture where the entire operating system is working in kernel space. The monolithic model differs from other operating system architectures, such as the microkernel architecture, in that it alone defines a high-level virtual interface over computer hardware.

A set of primitives or system calls implement all operating system services such as process management, concurrency, and memory management. Device drivers can be added to the kernel as modules.

Advantages of Monolithic Kernel

Here are the following advantages of a monolithic kernel, such as:

- The execution of the monolithic kernel is quite fast as the services such as memory management, file management, process scheduling, etc., are implemented under the same address space.
- A process runs completely in single address space in the monolithic kernel.

- The monolithic kernel is a static single binary file.

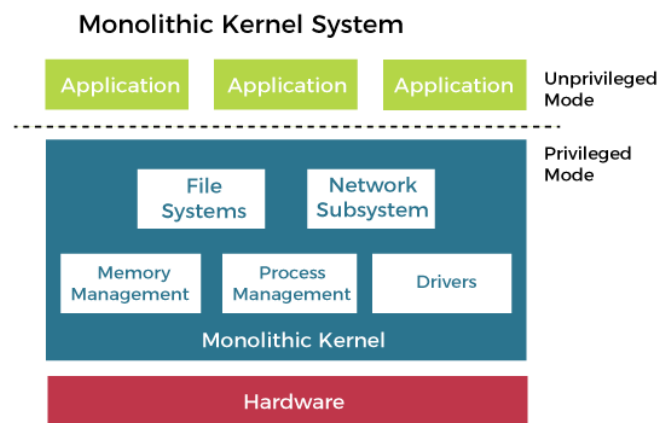
Disadvantages of Monolithic Kernel

Here are some disadvantages of the monolithic kernel, such as:

- If any service fails in the monolithic kernel, it leads to the failure of the entire system.
- The entire operating system needs to be modified by the user to add any new service.

Monolithic System Architecture:

A monolithic design of the operating system architecture makes no special accommodation for the special nature of the operating system. Although the design follows the separation of concerns, no attempt is made to restrict the privileges granted to the individual parts of the operating system. The entire operating system executes with maximum privileges. The communication overhead inside the monolithic operating system is the same as that of any other software, considered relatively low.



Hybrid structure:

All the architectures discussed so far have their own advantages and disadvantages. Monolithic systems are quite fast but their expansion is very difficult. Layered structure gives an efficient division of functionalities but if the number of layers is very high, it is difficult to manage the system. Microkernel architecture is quite efficient in isolating the core functionalities within the microkernel but the other services which are outside the kernel are not properly integrated.

And hence the idea was to combine the best functionalities of all these approaches and hence this design is termed as the hybrid structured operating system.

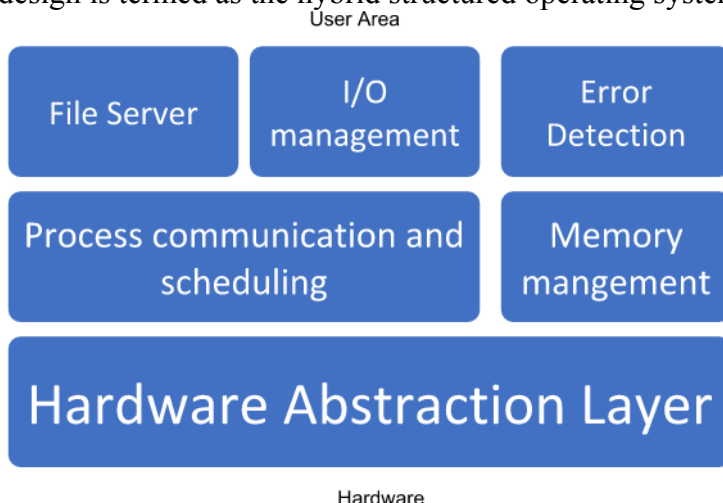


Fig. Hybrid architecture of operating system

It contains three layers:

1. **Hardware abstraction layer:** It is the lowermost layer that acts as an interface between the kernel and hardware.
2. **Microkernel layer:** This layer is nothing but the microkernel only which comprises of the three basic functionalities i.e., CPU scheduling, memory management, Inter-Process Communication.
3. **Application layer:** This layer is in user area and acts as an interface between user space and microkernel layer. It comprises of the remaining functionalities like file server, error detection, I/O device management etc.

In this way, the modular approach of microkernel structure and the layered approach both are restored, keeping the no. of layers easy to manage.

So, hybrid approach is highly useful and is largely used in present-day operating systems.

Most of the Mach operating systems run on this hybrid architecture only.

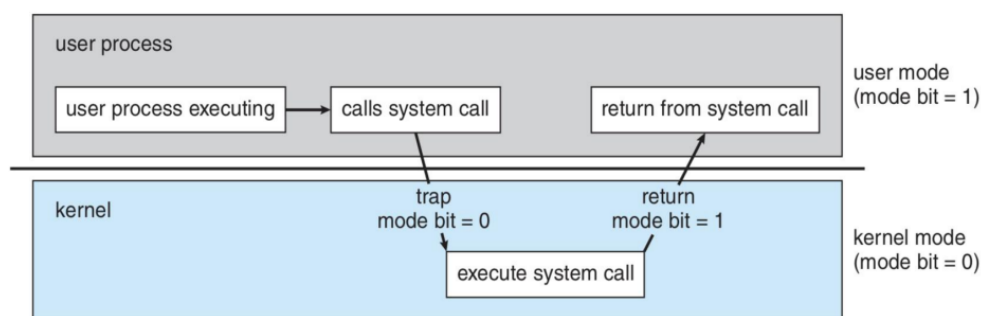
Advantages:

1. Easy to manage due to layered approach.
2. Number of layers is not very high.
3. Kernel is small and isolated.
4. Improved security and protection

Operating System Operations:

The following are different types of operations performed by each operating system.

• **Dual-Mode Operation:** Every operating system executes applications in two different modes: user mode and kernel mode (also called supervisor mode, system mode, or privileged mode). A bit, called the mode bit, is added to the hardware of the computer to indicate the current mode: kernel (0) or user (1). When the computer system is executing on behalf of a user application, the system is in user mode. However, when a user application requests a service from the operating system (via a system call), it must transition from user to kernel mode to fulfill the request, which is shown in the following diagram.



At system boot time, the hardware starts in kernel mode. The operating system is then loaded and starts user applications in user mode. Whenever a trap or interrupt occurs, the hardware switches from user mode to kernel mode (that is, changes the state of the mode bit to 0). Thus, whenever the operating system gains control of the computer, it is in kernel mode. The system always switches to user mode (by setting the mode bit to 1) before passing control to a user program. The hardware

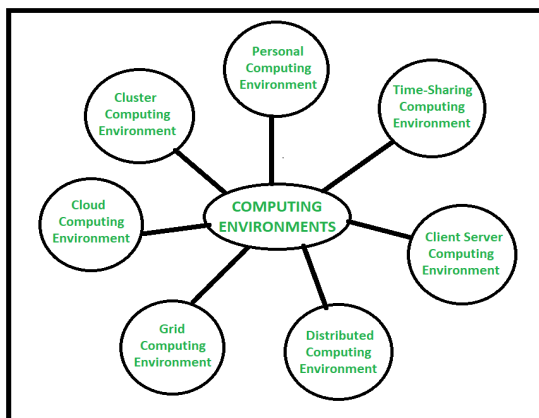
allows privileged instructions to be executed only in kernel mode. If an attempt is made to execute a privileged instruction in user mode, the hardware does not execute the instruction but rather treats it as illegal and traps it to the operating system.

- **Timer:** A timer can be set to interrupt the computer after a specified period. The period may be fixed (for example, 1/60 second) or variable (for example, from 1 millisecond to 1 second). Timer prevents a user program from getting stuck in an infinite loop or not calling system services and never returning control to the operating system.

Computing environments:

When a problem is solved by the computer, during that computer uses many devices, arranged in different ways and which work together to solve problems. This constitutes a computing environment where various number of computer devices arranged in different ways to solve different types of problems in different ways. In different computing environments computer devices are arranged in different ways and they exchange information in between them to process and solve problem. One computing environment consists of many computers' other computational devices, software and networks that to support processing and sharing information and solving task. Based on the organization of different computer devices and communication processes there exists multiple types of **computing environments**. Now lets know about different types of computing environments.

Types of Computing Environments : There are the various types of computing environments. They are :

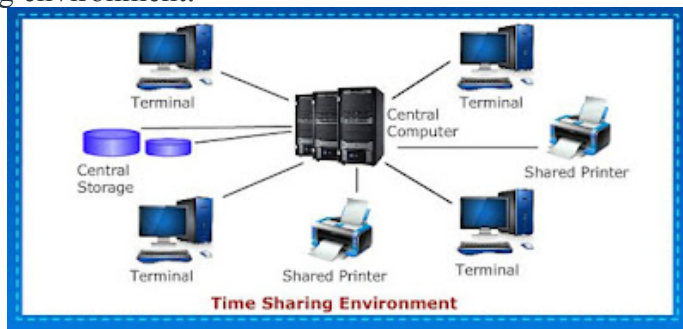


Computing Environments Types

1. **Personal Computing Environment :** In personal computing environment there is a stand-alone machine. Complete program resides on computer and executed there. Different stand-alone machines that constitute a personal computing environment are laptops, mobiles, printers, computer systems, scanners etc. That we use at our homes and offices.



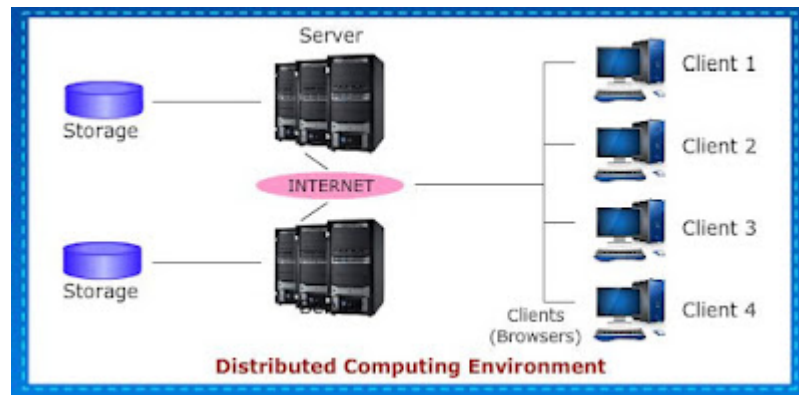
2. **Time-Sharing Computing Environment** : In Time Sharing Computing Environment multiple users share system simultaneously. Different users (different processes) are allotted different time slice and processor switches rapidly among users according to it. For example, student listening to music while coding something in an IDE. Windows 95 and later versions, Unix, IOS, Linux operating systems are the examples of this time sharing computing environment.



3. **Client Server Computing Environment** : In client server computing environment two machines are involved i.e., client machine and server machine, sometime same machine also serve as client and server. In this computing environment client requests resource/service and server provides that respective resource/service. A server can provide service to multiple clients at a time and here mainly communication happens through computer network.



4. **Distributed Computing Environment** : In a distributed computing environment multiple nodes are connected together using network but physically they are separated. A single task is performed by different functional units of different nodes of distributed unit. Here different programs of an application run simultaneously on different nodes, and communication happens in between different nodes of this system over network to solve task.



5. **Grid Computing Environment :** In grid computing environment, multiple computers from different locations work on a single problem. In this system, a set of computer nodes running in a cluster jointly perform a given task by applying resources of multiple computers/nodes. It is a network of computing environment where several scattered resources provide a running environment for a single task.
6. **Cloud Computing Environment :** In a cloud computing environment, on-demand availability of computer system resources like processing and storage are available. Here, computing is not done in individual technology or computer; rather, it is computed in a cloud of computers where all required resources are provided by a cloud vendor. This environment is primarily comprised of three services.
7. **Cluster Computing Environment :** In a cluster computing environment, a cluster performs a task where a cluster is a set of loosely or tightly connected computers that work together. It is viewed as a single system and performs tasks parallelly; that's why it is also similar to a parallel computing environment. Cluster-aware applications are especially used in a cluster computing environment.

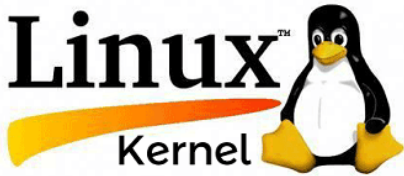
Open-Source Operating Systems

The term "**open source**" refers to computer software or applications where the owners or copyright holders enable the users or third parties to use, see, and edit the product's source code. The source code of an open-source OS is publicly visible and editable. The usual operating systems such as Apple's iOS, Microsoft's Windows, and Apple's Mac OS are closed operating systems. Open-Source Software is licensed in such a way that it is permissible to produce as many copies as you want and to use them wherever you like. It generally uses fewer resources than its commercial counterpart because it lacks any code for licensing, promoting other products, authentication, attaching advertisements, etc.

The open-source operating system allows the use of code that is freely distributed and available to anyone and for commercial purposes. Being an open-source application or program, the program source code of an open-source OS is available. The user may modify or change those codes and develop new applications according to the user requirement. Some basic examples of the open-source operating systems are **Linux, Open Solaris, Free RTOS, Open BDS, Free BSD, Minix**, etc.

Most of the open-source operating systems are Linux based. Some of the best open-source operating systems are as follows:

1. Linux Kernel



Linux kernel was developed by Linus Torvalds. It offers the essential functions required for an operating system, such as data cancellation, memory processing, and interactions with computer hardware. It is open-source software, and many developers researched the source code and produced a plethora of helpful plug-ins and operating systems to meet their requirements.

2. Linux Lite



Linux Lite is another free and open-source operating system that can run on lower-end hardware. It is a lightweight operating system designed to help users who are unfamiliar with Linux-based operating systems. The operating system includes all of the required programs, capabilities, tools, and desktops. It has a minimal interface and is entirely based on the Ubuntu system. In the last five years, the operating system has been stable and has received regular updates. It is efficiently functional soon after installation. After installation, users are not required to install any further drivers. If you want a lightweight open-source operating system on your PC, go with Linux Lite.

3. Linux mint



Linux Mint is a powerful Linux-based operating system that exudes modernity and power. It is simple to use and includes complete multimedia capabilities, making it a user-friendly open-source operating system. It is an Ubuntu-based distribution that is popular among both beginners and experts. It is built on the Debian platform and includes one of the most powerful software managers. It is more stable and has better visual aesthetics than Ubuntu.

4. Fedora



Fedora is another popular Linux-based operating system, and it is widely considered the best open-source operating system after Ubuntu. It is an RPM-based general-purpose operating system that is supported by Red Hat and built by the Fedora Project community. Its purpose is to develop and share cutting-edge open-source technology for free. As a result, Fedora developers prefer to make upstream improvements rather than create fixes specifically for Fedora. Fedora developers' updates are available to all Linux distributions.

5. React OS



ReactOS is another free and open-source operating system that has nearly 1 million downloads in over **100** countries. This community-based OS may run Windows apps, making it an excellent alternative to the Windows operating system. Although ReactOS is still growing, users, who love highly customizable operating systems, can select ReactOS. However, the operating system is developer-focused.

6. Solus



Solus is a free and open-source operating system for your desktop computer. It's a new operating system from the Linux family, released in **2012**. More than **6000** registered users are currently using the software. VLC, XChat, Transmission, Thunderbird, OpenShot Video Editor, Firefox, Budgie desktop environment, and LibreOffice Suite are all included with Solus. The most recent version of Solus, **Solus 3**, was released in **August 2017**.

7. Chrome OS



Chrome OS

Chrome OS is a partly open-source operating system with various attractive features. It's a part of the Chromium and Linux families, with features including better security, compatibility for supported Android and Chrome apps, Aura windows manager, Google cloud print, integrated media player, virtual desktop access, and cloud-based management. The only issue with the operating system is that it only supports Nexus devices or its hardware. As a result, if you're a Google fan, you'll love Chrome OS on a Chromebook.

Advantages and Disadvantages of Open-Source Operating System

Various advantages and disadvantages of the open-source operating system are as follows:

Advantages

1. Reliable and efficient

The open-source operating systems are most reliable and efficient. Thousands of eyes monitor these because the source code is public. As a result, if there are any bugs or errors, they are fixed by the best developers worldwide.

2. Cost-efficient

Most of the open-source operating systems are free. And some of them are far less expensive than commercially closed products.

3. Flexibility

The great advantage is you may customize it as per your requirement. And there is creative freedom.

Disadvantages

1. Complicated

It is not as user-friendly as the ones that are closed. To use this software, you must have a basic understanding of technology.

2. Security risk

Despite the defects having been detected, there is a risk of assaults because the attackers have access to the source code.

3. No support

If you run across an issue, there is no customer support available to assist you.

Operating System Services:

The operating system provides the programming environment in which a programmer works on a computer system. The user program requests various resources through the operating system. The operating system gives several services to utility programmers and users. Applications access these services through application programming interfaces or system calls. By invoking those interfaces, the application can request a service from the operating system, pass parameters, and acquire the operation outcomes.

Following are the services provided by an operating system -

- Program execution

B. Keerthana, Ass.Prof, CSE department

- Control Input/output devices
- Program creation
- Error Detection and Response
- Accounting
- Security and Protection
- File Management
- Communication

Program execution

To execute a program, several tasks need to be performed. Both the instructions and data must be loaded into the main memory. In addition, input-output devices and files should be initialized, and other resources must be prepared. The Operating structures handle these kinds of tasks. The user now no longer should fear the reminiscence allocation or multitasking or anything.

Control Input/output devices

As there are numerous types of I/O devices within the computer system, and each I/O device calls for its own precise set of instructions for the operation. The Operating System hides that info with the aid of presenting a uniform interface. Thus, it is convenient for programmers to access such devices easily.

Program Creation

The Operating system offers the structures and tools, including editors and debuggers, to help the programmer create, modify, and debugging programs.

Error Detection and Response

An Error in a device may also cause malfunctioning of the entire device. These include hardware and software errors such as device failure, memory error, division by zero, attempts to access forbidden memory locations, etc. To avoid error, the operating system monitors the system for detecting errors and takes suitable action with at least impact on running applications.

While working with computers, errors may occur quite often. Errors may occur in the:

- **Input/ Output devices:** For example, connection failure in the network, lack of paper in the printer, etc.
- **User program:** For example: attempt to access illegal memory locations, divide by zero, use too much CPU time, etc.
- **Memory hardware:** For example, Memory error, the memory becomes full, etc.

To handle these errors and other types of possible errors, the operating system takes appropriate action and generates messages to ensure correct and consistent computing.

Accounting

An Operating device collects utilization records for numerous assets and tracks the overall performance parameters and responsive time to enhance overall performance. These personal records are beneficial for additional upgrades and tuning the device to enhance overall performance.

Security and Protection

Operating device affords safety to the statistics and packages of a person and protects any interference from unauthorized users. The safety feature counters threats, which are published via way of individuals out of doors the manage of the running device.

For Example:

When a user downloads something from the internet, that program may contain malicious code that may harm the already existing programs. The operating system ensures that proper checks are applied while downloading such programs.

If one computer system is shared amongst a couple of users, then the various processes must be protected from another intrusion. For this, the operating system provides various mechanisms that allow only those processes to use resources that have gained proper authorization from the operating system. The mechanism may include providing unique users ids and passwords to each user.

File management

Computers keep data and information on secondary storage devices like magnetic tape, magnetic disk, optical disk, etc. Each storage media has its capabilities like speed, capacity, data transfer rate, and data access methods.

For file management, the operating system must know the types of different files and the characteristics of different storage devices. It has to offer the proportion and safety mechanism of documents additionally.

Communication

The operating system manages the exchange of data and programs among different computers connected over a network. This communication is accomplished using message passing and shared memory.

System calls:

A system call is a method for a computer program to request a service from the kernel of the operating system on which it is running. A system call is a method of interacting with the operating system via programs. A system call is a request from computer software to an operating system's kernel.

The **Application Program Interface (API)** connects the operating system's functions to user programs. It acts as a link between the operating system and a process, allowing user-level programs to request operating system services. The kernel system can only be accessed using system calls. System calls are required for any programs that use resources.

How are system calls made?

When a computer software needs to access the operating system's kernel, it makes a system call. The system call uses an API to expose the operating system's services to user programs. It is the only method to access the kernel system. All programs or processes that require resources for execution must use system calls, as they serve as an interface between the operating system and user programs. Below are some examples of how a system call varies from a user function.

1. A system call function may create and use kernel processes to execute the asynchronous processing.
2. A system call has greater authority than a standard subroutine. A system call with kernel-mode privilege executes in the kernel protection domain.
3. System calls are not permitted to use shared libraries or any symbols that are not present in the kernel protection domain.
4. The code and data for system calls are stored in global kernel memory.

Why do you need system calls in Operating System?

There are various situations where you must require system calls in the operating system. Following of the situations are as follows:

1. It is must require when a file system wants to create or delete a file.
2. Network connections require the system calls to sending and receiving data packets.
3. If you want to read or write a file, you need to system calls.

4. If you want to access hardware devices, including a printer, scanner, you need a system call.
5. System calls are used to create and manage new processes.

How System Calls Work

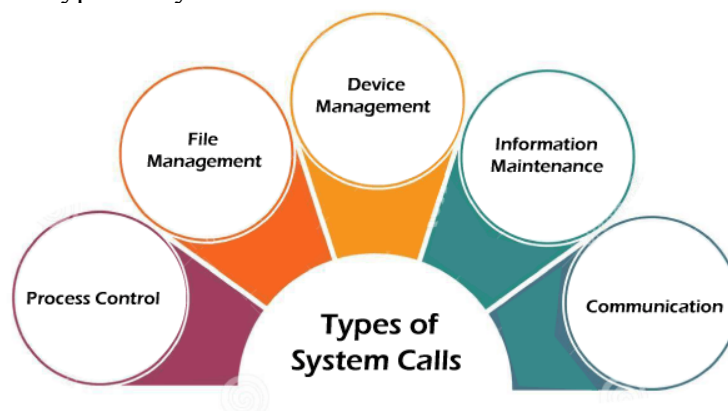
The Applications run in an area of memory known as user space. A system call connects to the operating system's kernel, which executes in kernel space. When an application creates a system call, it must first obtain permission from the kernel. It achieves this using an interrupt request, which pauses the current process and transfers control to the kernel.

If the request is permitted, the kernel performs the requested action, like creating or deleting a file. As input, the application receives the kernel's output. The application resumes the procedure after the input is received. When the operation is finished, the kernel returns the results to the application and then moves data from kernel space to user space in memory.

A simple system call may take few nanoseconds to provide the result, like retrieving the system date and time. A more complicated system call, such as connecting to a network device, may take a few seconds. Most operating systems launch a distinct kernel thread for each system call to avoid bottlenecks. Modern operating systems are multi-threaded, which means they can handle various system calls at the same time.

Types of System Calls

There are commonly five types of system calls. These are as follows:



1. **Process Control**
2. **File Management**
3. **Device Management**
4. **Information Maintenance**
5. **Communication**

Now, you will learn about all the different types of system calls one-by-one.

Process Control

Process control is the system call that is used to direct the processes. Some process control examples include creating, load, abort, end, execute, process, terminate the process, etc.

File Management

File management is a system call that is used to handle the files. Some file management examples include creating files, delete files, open, close, read, write, etc.

Device Management

Device management is a system call that is used to deal with devices. Some examples of device management include read, device, write, get device attributes, release device, etc.

Information Maintenance

Information maintenance is a system call that is used to maintain information. There are some examples of information maintenance, including getting system data, set time or date, get time or date, set system data, etc.

Communication

Communication is a system call that is used for communication. There are some examples of communication, including create, delete communication connections, send, receive messages, etc.

Examples of Windows and Unix system calls:

There are various examples of Windows and Unix system calls. These are as listed below in the table:

Process	Windows	Unix
Process Control	CreateProcess(), ExitProcess() WaitForSingleObject()	Fork(),Exit(),Wait()
File Manipulation	CreateFile(),ReadFile(),WriteFile() CloseHandle()	Open(),Read(),Write() Close()
Device Management	SetConsoleMode() ReadConsole() WriteConsole()	Ioctl(),Read(),Write()
Information Maintenance	GetCurrentProcessID() SetTimer(), Sleep()	Getpid(),Alarm() Sleep()
Communication	CreatePipe(),CreateFileMapping() MapViewOfFile()	Pipe(),Shmget(),Mmap()
Protection	SetFileSecurity(),InitializeSecurityDescriptor() SetSecurityDescriptorgroup()	Chmod() Umask(),Chown()

Here, you will learn about some methods briefly:

open()

The **open()** system call allows you to access a file on a file system. It allocates resources to the file and provides a handle that the process may refer to. Many processes can open a file at once or by a single process only. It's all based on the file system and structure.

read()

It is used to obtain data from a file on the file system. It accepts three arguments in general:

- A file descriptor.
- A buffer to store read data.

- The number of bytes to read from the file.

The file descriptor of the file to be read could be used to identify it and open it using **open()** before reading.

wait()

In some systems, a process may have to wait for another process to complete its execution before proceeding. When a parent process makes a child process, the parent process execution is suspended until the child process is finished. The **wait()** system call is used to suspend the parent process. Once the child process has completed its execution, control is returned to the parent process.

write()

It is used to write data from a user buffer to a device like a file. This system call is one way for a program to generate data. It takes three arguments in general:

- A file descriptor.
- A pointer to the buffer in which data is saved.
- The number of bytes to be written from the buffer.

fork()

Processes generate clones of themselves using the **fork()** system call. It is one of the most common ways to create processes in operating systems. When a parent process spawns a child process, execution of the parent process is interrupted until the child process completes. Once the child process has completed its execution, control is returned to the parent process.

close()

It is used to end file system access. When this system call is invoked, it signifies that the program no longer requires the file, and the buffers are flushed, the file information is altered, and the file resources are de-allocated as a result.

exec()

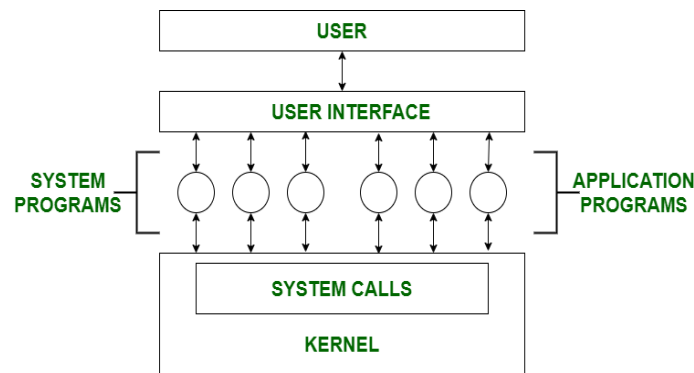
When an executable file replaces an earlier executable file in an already executing process, this system function is invoked. As a new process is not built, the old process identification stays, but the new process replaces data, stack, data, head, etc.

exit()

The **exit()** is a system call that is used to end program execution. This call indicates that the thread execution is complete, which is especially useful in multi-threaded environments. The operating system reclaims resources spent by the process following the use of the **exit()** system function.

System Programs:

System Programming can be defined as the act of building Systems Software using System Programming Languages. It provides an environment to develop and execute the programs. According to Computer Hierarchy, one which comes at last is Hardware. Then it is Operating System, System Programs, and finally Application Programs. Program Development and Execution can be done conveniently in System Programs. Some of the System Programs are simply user interfaces, others are complex. It traditionally lies between the user interface and system calls.



So here, the user can only view up-to-the System Programs he can't see System Calls. System Programs can be divided into these categories :

1. **File Management** – A file is a collection of specific information stored in the memory of a computer system. File management is defined as the process of manipulating files in the computer system, its management includes the process of **creating, modifying and deleting files**.
 - It helps to **create** new files in the computer system and placing them at specific locations.
 - It helps in easily and quickly **locating these files** in the computer system.
 - It makes the process of **sharing files** among different users very easy and user-friendly.
 - It helps to **store files** in separate folders known as directories.
 - These directories help users to **search files** quickly or to manage files according to their types of uses.
 - It helps users to modify the data of files or to **modify the name of files** in directories.
2. **Status Information** – Information like **date, time amount of available memory, or disk space** is asked by some users. Others providing **detailed performance, logging, and debugging information which is more complex**. All this information is formatted and displayed on output devices or printed. Terminal or other output devices or files or a window of GUI is used for showing the output of programs.
3. **File Modification** – For **modifying the contents of files** we use this. For Files stored on disks or other storage devices, we used different types of editors. For searching contents of files or perform transformations of files we use special commands.
4. **Programming-Language support** – For common programming languages, we use **Compilers, Assemblers, Debuggers, and interpreters** which are already provided to users. It provides all support to users. We can run any programming language. All languages of importance are already provided.
5. **Program Loading and Execution** – When the program is ready after Assembling and compilation, it must be loaded into memory for execution. **A loader is part of an operating system that is responsible for loading programs and libraries**. It is one of the essential stages for starting a program. Loaders, relocatable loaders, linkage editors, and Overlay loaders are provided by the system.
6. **Communications** – **Virtual connections** among processes, users, and computer systems are provided by programs. Users can send messages to another user on their screen, User can send e-mail, **browsing on web pages, remote login, the transformation of files** from one user to another.

Operating System Debugging And System Boot:

Debugging is the process of **finding the problems in a computer system and solving them**. There are many different ways in which operating systems perform debugging. Some of these are –

Log Files:

The log files **record all the events** that occur in an operating system. This is done by writing all the messages into a log file. There are different types of log files. Some of these are given as follows –

a) Event Logs

These stores the records of **all the events that occur in the execution** of a system. This is done so that the activities of all the events can be understood to diagnose problems.

b) Transaction Logs

The transaction logs **store the changes to the data** so that the system can recover from crashes and other errors. These logs are readable by a human.

c) Message Logs

These logs **store both the public and private messages** between the users. They are mostly plain text files, but in some cases they may be HTML files.

Core Dump Files

The core dump files contain the **memory address space of a process that terminates unexpectedly**. The creation of the **core dump is triggered in response to program crashes by the kernel**. The core dump files are used by the developers to find the program's state at the time of its termination so that they can find out why the termination occurred.

The automatic creation of the core dump files can be disabled by the users. This may be done to improve performance, clear disk space or increase security.

Crash Dump Files

In the event of a total system failure, the information about the state of the operating system is captured in crash dump files. There are three types of dump that can be captured when a system crashes. These are –

Trace Listings

The trace listing **record information about a program execution using logging**. This information is **used by programmers for debugging**. System administrators and technical personnel can use the trace listings to find the common problems with software using software monitoring tools.

System booting:

Bootting is the process of starting a computer. It can be initiated by hardware such as a button press or by a software command. After it is switched on, a CPU has no software in its main memory, so some processes must load software into memory before execution. This may be done by hardware or firmware in the CPU or by a separate processor in the computer system.

Types of Booting



1. **Cold Booting:** When the **computer starts for the first time** or is in a **shut-down state and switch on the power button** to start the system, this type of process to start the computer is called cold booting. During cold booting, the system will read all the **instructions from the ROM (BIOS)** and the **Operating System will be automatically get loaded into the system**. This booting takes **more time than Hot or Warm Booting**.
2. **Warm Booting:** Warm or Hot Booting process is when **computer systems come to no response or hang state**, and then **the system is allowed to restart during on condition**. It is also referred to as **rebooting**. There are many reasons for this state, and the only solution is to reboot the computer. Rebooting may be **required when we install new software or hardware**. The system requires a reboot to set software or hardware **configuration changes**, or sometimes systems may behave abnormally or may not respond properly. In such a case, the system has to be a **force restart**. Most commonly **Ctrl+Alt+Del button is used to reboot the system**. Else, in some systems, the **external reset button may be available** to reboot the system.

Bootting Process in Operating System

When our computer is switched on, it can be started by hardware such as a button press, or by software command, a computer's central processing unit (CPU) has no software in its main memory, there is some process which must load software into main memory before it can be executed. Below are the six steps to describe the boot process in the operating system, such as:



Step 1: Once the computer system is turned on, **BIOS** (Basic Input /Output System) performs a series of activities or functionality tests on programs stored in ROM, called on **POST** (Power-on Self Test) that checks to see whether peripherals in the system are in perfect order or not.

Step 2: After the BIOS is done with pre-boot activities or functionality test, it read bootable sequence from **CMOS** (Common Metal Oxide Semiconductor) and looks for master boot record in the first physical sector of the bootable disk as per boot device sequence specified in **CMOS**. For example, if the boot device sequence is:

- Floppy Disk
- Hard Disk
- CDROM

Step 3: After this, the master boot record will search first in a **floppy disk drive**. If not found, then the hard disk drive will search for the master boot record. But if the master boot record is not even present on the hard disk, then the CDROM drive will search. If the system cannot read the master boot record from any of these sources, ROM displays **"No Boot device found"** and halted the system. On finding the master boot record from a particular bootable disk drive, the operating system loader, also called Bootstrap loader, is loaded from the boot sector of that bootable drive into memory. A bootstrap loader is a special program that is present in the boot sector of a bootable drive.

Step 4: The bootstrap loader first loads the **IO.SYS** file. After this, **MSDOS.SYS** file is loaded, which is the core file of the DOS operating system.

Step 5: After this, *MSDOS.SYS* file searches to find Command Interpreter in *CONFIG.SYS* file, and when it finds, it loads into memory. If no Command Interpreter is specified in the *CONFIG.SYS* file, the *COMMAND.COM* file is loaded as the default Command Interpreter of the DOS operating system.

Step 6: The last file is to be loaded and executed is the *AUTOEXEC.BAT* file that contains a sequence of DOS commands. After this, the prompt is displayed. We can see the drive letter of bootable drive displayed on the computer system, which indicates that the operating system has been successfully on the system from that drive.

What is Dual Booting

When two operating systems are installed on the computer system, then it is called dual booting. Multiple operating systems can be installed on such a system. But to know which operating system is to boot, a boot loader that understands multiple file systems and multiple operating systems can occupy the boot space.

Once loaded, it can boot one of the operating systems available on the disk. The disk can have multiple partitions, each containing a different type of operating system. When a computer system turns on, a boot manager program displays a menu, allowing the user to choose the operating system to use.